

LP V Assignment No 2

Name : Bhure Sangmeshwar Dattatray

Roll No : 07

```
In [5]: # =====
# IMDB Binary Classification (No Matplotlib Version)
# =====

import numpy as np
import warnings
warnings.filterwarnings("ignore")

import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
from tensorflow.keras.datasets import imdb
from tensorflow.keras.callbacks import EarlyStopping

from sklearn.metrics import classification_report

print("TensorFlow version:", tf.__version__)

# Parameters
NUM_WORDS = 10000
MAX_LEN = 256

# Load dataset
(x_train_raw, y_train), (x_test_raw, y_test) = imdb.load_data(num_w

# Multi-hot encoding
def multi_hot(sequences):
    result = np.zeros((len(sequences), NUM_WORDS))
    for i, seq in enumerate(sequences):
        result[i, seq] = 1
    return result

x_train = multi_hot(x_train_raw)
x_test = multi_hot(x_test_raw)

# Split validation
val_size = int(0.2 * len(x_train))
x_val, x_train = x_train[:val_size], x_train[val_size:]
y_val, y_train = y_train[:val_size], y_train[val_size:]

# Model
model = keras.Sequential([
    layers.Input(shape=(NUM_WORDS,)),
    layers.Dense(64, activation='relu'),
    layers.Dropout(0.4),
    layers.Dense(64, activation='relu'),
    layers.Dense(1, activation='sigmoid')
])
```

```
model.compile(optimizer='adam',
              loss='binary_crossentropy',
              metrics=['accuracy'])

# Train
history = model.fit(
    x_train, y_train,
    epochs=10,
    batch_size=512,
    validation_data=(x_val, y_val),
    callbacks=[EarlyStopping(patience=2)]
)

# Evaluate
loss, acc = model.evaluate(x_test, y_test)
print("\nTest Accuracy:", acc)

# Predictions
y_pred = (model.predict(x_test) > 0.5).astype(int)

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

print("\nAssignment Completed Successfully ✅")
```

TensorFlow version: 2.21.0

Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/imdb.npz>

17464789/17464789  **3s 0us/step**

Epoch 1/10

40/40  **2s 21ms/step** - accuracy: 0.7821 - loss: 0.4791 - val_accuracy: 0.8826 - val_loss: 0.2899

Epoch 2/10

40/40  **1s 11ms/step** - accuracy: 0.9103 - loss: 0.2378 - val_accuracy: 0.8934 - val_loss: 0.2709

Epoch 3/10

40/40  **0s 9ms/step** - accuracy: 0.9416 - loss: 0.1621 - val_accuracy: 0.8922 - val_loss: 0.2854

Epoch 4/10

40/40  **0s 11ms/step** - accuracy: 0.9596 - loss: 0.1141 - val_accuracy: 0.8914 - val_loss: 0.3135

782/782  **1s 801us/step** - accuracy: 0.8764 - loss: 0.3432

Test Accuracy: 0.8763999938964844

782/782  **1s 640us/step**

Classification Report:

	precision	recall	f1-score	support
0	0.88	0.87	0.88	12500
1	0.87	0.88	0.88	12500
accuracy			0.88	25000
macro avg	0.88	0.88	0.88	25000
weighted avg	0.88	0.88	0.88	25000

Assignment Completed Successfully 

In []: